Atiyah–Macdonald, Chapter 7: Noetherian Rings

Solutions to the exercises

Conventions. Rings are commutative with 1. From the text I use: the equivalent forms of the Noetherian condition (ACC, maximal condition, every ideal finitely generated; 6.1, 6.2); quotients of Noetherian rings and finitely generated modules over Noetherian rings are Noetherian (6.5, 6.6, 7.1); localizations (7.3) and finitely generated algebras (7.7, Hilbert's basis theorem 7.5) over Noetherian rings are Noetherian; Prop. 7.8 and Prop. 7.9/Cor. 7.10 (a field which is a finitely generated k-algebra is a finite extension of k); an ideal of a Noetherian ring contains a power of its radical (7.14); irreducible ideals are primary (7.12) and the primes belonging to an ideal are the primes $r(\mathfrak{a} : x)$ (4.5, 7.17). From earlier chapters: Ch. 3, Ex. 16, 18, 21, 22 and Ch. 4, Ex. 20–22. A topological space is *Noetherian* if its closed sets satisfy the descending chain condition (equivalently every subspace is quasi-compact; Ch. 6, Ex. 5); $\mathrm{Spec}(A)$ is a Noetherian space if A is a Noetherian ring (a descending chain $Y_1 \supseteq Y_2 \supseteq \cdots$ of closed sets gives an ascending chain of ideals $\bigcap_{\mathfrak{p}\in Y_n} \mathfrak{p}$, and $Y_n = V(\bigcap_{\mathfrak{p}\in Y_n} \mathfrak{p})$). In Exercises 23 and 24 I use the key lemma of Ch. 5, Ex. 21: *if $A \subseteq B$ are integral domains with B finitely generated over A, there is $u \neq 0$ in A such that every homomorphism of A into an algebraically closed field Ω with $f(u) \neq 0$ extends to a homomorphism $B \to \Omega$.*

This chapter has no numbered in-text exercises; the book lists 27 end-of-chapter exercises and all of them are solved here (Exercises 26–27 on Grothendieck groups were added after an independent review).

End-of-chapter exercises

Exercise 1

> **Problem.** Let A be a non-Noetherian ring and let Σ be the set of ideals in A which are not finitely generated. Show that Σ has maximal elements and that the maximal elements of Σ are prime ideals.
>
> [Let $\mathfrak{a}$ be a maximal element of Σ, and suppose that there exist $x, y \in A$ such that $x \notin \mathfrak{a}$ and $y \notin \mathfrak{a}$ and $xy \in \mathfrak{a}$. Show that there exists a finitely generated ideal $\mathfrak{a}_0 \subseteq \mathfrak{a}$ such that $\mathfrak{a}_0 + (x) = \mathfrak{a} + (x)$, and that $\mathfrak{a} = \mathfrak{a}_0 + x \cdot (\mathfrak{a} : x)$. Since $(\mathfrak{a} : x)$ strictly contains $\mathfrak{a}$, it is finitely generated and therefore so is $\mathfrak{a}$.]
>
> Hence a ring in which every prime ideal is finitely generated is Noetherian (I. S. Cohen).

Solution. $\Sigma \neq \emptyset$ since A is not Noetherian. If $(\mathfrak{a}_\lambda)$ is a chain in Σ and $\mathfrak{a} = \bigcup \mathfrak{a}_\lambda$ were finitely generated, its finitely many generators would lie in one $\mathfrak{a}_\lambda$, forcing $\mathfrak{a}_\lambda = \mathfrak{a}$ finitely generated, a contradiction. So $\mathfrak{a} \in \Sigma$ and by Zorn's lemma Σ has maximal elements.

Let $\mathfrak{a}$ be maximal in Σ. Then $\mathfrak{a} \neq (1)$. Suppose $xy \in \mathfrak{a}$ with $x, y \notin \mathfrak{a}$. The ideal $\mathfrak{a} + (x) \supsetneq \mathfrak{a}$ is not in Σ, so it is finitely generated, say by $a_1 + c_1 x, \ldots, a_n + c_n x$ with $a_i \in \mathfrak{a}$. Put $\mathfrak{a}_0 = (a_1, \ldots, a_n) \subseteq \mathfrak{a}$; then $\mathfrak{a} + (x) = \mathfrak{a}_0 + (x)$. We claim $\mathfrak{a} = \mathfrak{a}_0 + x(\mathfrak{a} : x)$. The inclusion $\supseteq$ is clear. If $z \in \mathfrak{a}$, then $z \in \mathfrak{a}_0 + (x)$, say $z = b + cx$ with $b \in \mathfrak{a}_0$; then $cx = z - b \in \mathfrak{a}$, so $c \in (\mathfrak{a} : x)$. Now $(\mathfrak{a} : x) \ni y$ and $y \notin \mathfrak{a}$, so $(\mathfrak{a} : x) \supsetneq \mathfrak{a}$, hence $(\mathfrak{a} : x) \notin \Sigma$ is finitely generated; then $\mathfrak{a} = \mathfrak{a}_0 + x(\mathfrak{a} : x)$ is finitely generated, a contradiction. So $\mathfrak{a}$ is prime.

If every prime of A were finitely generated, Σ would be empty (a maximal element of a non-empty Σ would be a non-finitely-generated prime), i.e. A is Noetherian. □

Exercise 2

Problem. Let A be a Noetherian ring and let $f = \sum_{n=0}^{\infty} a_n x^n \in A[[x]]$. Prove that f is nilpotent if and only if each a_n is nilpotent.

Solution. ($\Rightarrow$) is Ch. 1, Ex. 5(ii) (valid for every ring).

($\Leftarrow$) Let all $a_n \in \mathfrak{N}$, the nilradical of A. Since A is Noetherian, $\mathfrak{N}$ is nilpotent (7.15): $\mathfrak{N}^N = 0$ for some N. The coefficient of x^k in f^N is a sum of products $a_{i_1} \cdots a_{i_N} \in \mathfrak{N}^N = 0$. So $f^N = 0$. □

Exercise 3

Problem. Let $\mathfrak{a}$ be an irreducible ideal in a ring A. Then the following are equivalent:

(i) $\mathfrak{a}$ is primary;

(ii) for every multiplicatively closed subset S of A we have $(S^{-1}\mathfrak{a})^c = (\mathfrak{a} : x)$ for some $x \in S$;

(iii) the sequence $(\mathfrak{a} : x^n)$ is stationary, for every $x \in A$.

Solution. (As usual an irreducible ideal is proper.) Recall $(S^{-1}\mathfrak{a})^c = \bigcup_{s \in S} (\mathfrak{a} : s)$ (3.11).

(i)$\Rightarrow$(ii). Let $\mathfrak{a}$ be $\mathfrak{p}$-primary. If $S \cap \mathfrak{p} = \emptyset$, then $(S^{-1}\mathfrak{a})^c = \mathfrak{a} = (\mathfrak{a} : 1)$ by 4.8(ii), and $1 \in S$. If $s \in S \cap \mathfrak{p}$, then $s^n \in \mathfrak{a}$ for some n, and $s^n \in S$; $(S^{-1}\mathfrak{a})^c = (1) = (\mathfrak{a} : s^n)$ by 4.8(i).

(ii)$\Rightarrow$(iii). Take $S = \{x^n\}_{n \geq 0}$: $\bigcup_n (\mathfrak{a} : x^n) = (\mathfrak{a} : x^m)$ for some m, and since the chain $(\mathfrak{a} : x^n)$ is ascending, $(\mathfrak{a} : x^n) = (\mathfrak{a} : x^m)$ for all $n \geq m$.

(iii)$\Rightarrow$(i). Pass to $\bar{A} = A/\mathfrak{a}$, in which 0 is irreducible and $\mathrm{Ann}(\bar{x}^n)$ is stationary for all $\bar{x}$ (note $\mathrm{Ann}_{\bar{A}}(\bar{x}^n) = (\mathfrak{a} : x^n)/\mathfrak{a}$). Let $\bar{x}\bar{y} = 0$ with $\bar{y} \neq 0$; choose n with $\mathrm{Ann}(\bar{x}^n) = \mathrm{Ann}(\bar{x}^{n+1})$. Then $(\bar{x}^n) \cap (\bar{y}) = 0$: if $a \in (\bar{y})$, then $a\bar{x} = 0$; if also $a = b\bar{x}^n$, then $b\bar{x}^{n+1} = 0$, so $b \in \mathrm{Ann}(\bar{x}^{n+1}) = \mathrm{Ann}(\bar{x}^n)$ and $a = b\bar{x}^n = 0$. Since 0 is irreducible and $(\bar{y}) \neq 0$, we get $\bar{x}^n = 0$. Thus every zero-divisor of $\bar{A} \neq 0$ is nilpotent, i.e. $\mathfrak{a}$ is primary. □

Exercise 4

Problem. Which of the following rings are Noetherian?

(i) The ring of rational functions of z having no pole on the circle $|z| = 1$.

(ii) The ring of power series in z with a positive radius of convergence.

(iii) The ring of power series in z with an infinite radius of convergence.

(iv) The ring of polynomials in z whose first k derivatives vanish at the origin (k being a fixed integer).

(v) The ring of polynomials in z, w all of whose partial derivatives with respect to w vanish for $z = 0$.

In all cases the coefficients are complex numbers.

Solution. (i) *Noetherian.* A rational function p/q in lowest terms has no pole on $|z| = 1$ iff q has no zero there. So the ring is $S^{-1}\mathbf{C}[z]$ with $S = \{q : q(z) \neq 0 \text{ for } |z| = 1\}$, Noetherian by 7.3 since $\mathbf{C}[z]$ is.

(ii) *Noetherian.* Let R be the ring of convergent power series. A non-zero $f \in R$ has the form $f = z^n u$ with $u(0) \neq 0$, and $1/u$ is again a convergent power series near 0, so u is a unit of R. If $I \neq 0$ is an ideal and n is the smallest order of an element of I, then $z^n \in I$ (it is $u^{-1}f$ for an f of order n), so $I = (z^n)$. Every ideal is principal.

(iii) *Not Noetherian.* Let R be the ring of entire functions and $J_n = \{f \in R : f(k) = 0 \text{ for all integers } k \geq n\}$. Then $J_1 \subseteq J_2 \subseteq \cdots$. For $n \geq 2$, $f_n(z) = \sin(\pi z)/\prod_{k=1}^{n-1}(z-k)$ is entire (the zeros of the denominator are simple zeros of $\sin \pi z$), it vanishes at all integers $k \geq n$, but $f_n(n-1) \neq 0$ (the zero of $\sin \pi z$ at $n-1$ is cancelled, and the other factors of the denominator do not vanish there). So $f_n \in J_n \setminus J_{n-1}$ and the chain is strictly ascending.

(iv) *Noetherian.* The ring is $R = \{f \in \mathbf{C}[z] : f'(0) = \cdots = f^{(k)}(0) = 0\}$, the polynomials $a_0 + a_{k+1}z^{k+1} + a_{k+2}z^{k+2} + \cdots$. It is generated as a $\mathbf{C}$-algebra by $z^{k+1}, \ldots, z^{2k+1}$: every $m \geq k+1$ is either in $[k+1, 2k+1]$ or $m - (k+1) \geq k+1$, and one inducts. So R is a finitely generated $\mathbf{C}$-algebra, Noetherian by 7.7.

(v) *Not Noetherian.* Write $f = \sum_j f_j(z)w^j$. The m-th w-derivative at $z = 0$ is $\sum_{j \geq m} \frac{j!}{(j-m)!} f_j(0) w^{j-m}$, and these vanish for all $m \geq 1$ iff $f_j(0) = 0$ for all $j \geq 1$ (descend from the top j). So $R = \{f : f_j(0) = 0 \ \forall j \geq 1\}$. Put $I_n = (zw, zw^2, \ldots, zw^n)R$. Then $I_1 \subseteq I_2 \subseteq \cdots$ and $zw^{n+1} \notin I_n$: an element $\sum_{j=1}^{n} r_j z w^j$ with $r_j \in R$ has w^{n+1}-coefficient $z \sum_j [w^{n+1-j}] r_j$, and for $n+1-j \geq 1$ the coefficient $[w^{n+1-j}]r_j$ lies in $z\mathbf{C}[z]$; so the w^{n+1}-coefficient lies in $z^2\mathbf{C}[z]$ and cannot equal z. The chain is strictly ascending. □

Exercise 5

Problem. Let A be a Noetherian ring, B a finitely generated A-algebra, G a finite group of A-automorphisms of B, and B^G the set of all elements of B which are left fixed by every element of G. Show that B^G is a finitely generated A-algebra.

Solution. B^G is a subring containing the image of A. Let $B = A[x_1, \ldots, x_n]$ and $f_i(t) = \prod_{g \in G}(t - gx_i) \in B[t]$. Each $h \in G$ permutes the factors, so $f_i \in B^G[t]$, and f_i is monic with $f_i(x_i) = 0$. Let $C \subseteq B^G$ be the A-subalgebra generated by the coefficients of $f_1, \ldots, f_n$. C is a finitely generated algebra over the Noetherian ring A, hence Noetherian (7.7). Each x_i is integral over C and $B = C[x_1, \ldots, x_n]$, so B is a finitely generated C-module (5.2). Since C

is Noetherian, B is a Noetherian C-module (6.5), so its submodule B^G is a finitely generated C-module (6.2). Hence B^G is a finitely generated C-algebra, and as C is a finitely generated A-algebra, B^G is a finitely generated A-algebra. □

Exercise 6

Problem. If a finitely generated ring K is a field, it is a finite field.

[If K has characteristic 0, we have $\mathbf{Z} \subset \mathbf{Q} \subseteq K$. Since K is finitely generated over $\mathbf{Z}$ it is finitely generated over $\mathbf{Q}$, hence by (7.9) is a finitely generated $\mathbf{Q}$-module. Now apply (7.8) to obtain a contradiction. Hence K is of characteristic $p > 0$, hence is finitely generated as a $\mathbf{Z}/(p)$-algebra. Use (7.9) to complete the proof.]

Solution. Suppose $\operatorname{char} K = 0$. Then $\mathbf{Z} \subset \mathbf{Q} \subseteq K$, and $K = \mathbf{Z}[a_1, \ldots, a_m] \subseteq \mathbf{Q}[a_1, \ldots, a_m] \subseteq K$, so K is a finitely generated $\mathbf{Q}$-algebra and a field; by 7.9 it is a finite extension of $\mathbf{Q}$, i.e. a finitely generated $\mathbf{Q}$-module. Apply 7.8 to $\mathbf{Z} \subseteq \mathbf{Q} \subseteq K$ ($\mathbf{Z}$ Noetherian, K a finitely generated $\mathbf{Z}$-algebra and a finitely generated $\mathbf{Q}$-module): $\mathbf{Q}$ is a finitely generated $\mathbf{Z}$-algebra, $\mathbf{Q} = \mathbf{Z}[u_1/v_1, \ldots, u_s/v_s]$. Every element of this ring has a denominator dividing a power of $v = v_1 \cdots v_s$. Choose a prime $p \nmid v$ (there are infinitely many); then $1/p$ is not in the ring, a contradiction.

So $\operatorname{char} K = p > 0$, and K is a finitely generated algebra over its prime field $\mathbf{F}_p = \mathbf{Z}/(p)$ (generators as a ring). By 7.9, K is a finite extension of $\mathbf{F}_p$, hence a finite-dimensional $\mathbf{F}_p$-vector space, hence a finite field. □

Exercise 7

Problem. Let X be an affine algebraic variety given by a family of equations $f_\alpha(t_1, \ldots, t_n) = 0$ ($\alpha \in I$) (Chapter 1, Exercise 27). Show that there exists a finite subset I_0 of I such that X is given by the equations $f_\alpha(t_1, \ldots, t_n) = 0$ for $\alpha \in I_0$.

Solution. Let $\mathfrak{a} \subseteq k[t_1, \ldots, t_n]$ be the ideal generated by all f_α. By Hilbert's basis theorem (7.6) it is finitely generated, say by $g_1, \ldots, g_m$. Each g_j is a finite combination of the f_α, so there is a finite $I_0 \subseteq I$ with $g_1, \ldots, g_m \in (f_\alpha : \alpha \in I_0)$; hence $\mathfrak{a} = (f_\alpha : \alpha \in I_0)$. A point is a common zero of all f_α iff it is a common zero of $\mathfrak{a}$ iff it is a common zero of the f_α, $\alpha \in I_0$. □

Exercise 8

Problem. If $A[x]$ is Noetherian, is A necessarily Noetherian?

Solution. Yes. $A \cong A[x]/(x)$ is a homomorphic image of $A[x]$ (via $x \mapsto 0$), hence Noetherian by 7.1. □

Exercise 9

Problem. Let A be a ring such that

(1) for each maximal ideal $\mathfrak{m}$ of A, the local ring $A_\mathfrak{m}$ is Noetherian;

(2) for each $x \neq 0$ in A, the set of maximal ideals of A which contain x is finite.

Show that A is Noetherian.

[Let $\mathfrak{a} \neq 0$ be an ideal in A. Let $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ be the maximal ideals which contain $\mathfrak{a}$. Choose $x_0 \neq 0$ in $\mathfrak{a}$ and let $\mathfrak{m}_1, \ldots, \mathfrak{m}_{r+s}$ be the maximal ideals which contain x_0. Since $\mathfrak{m}_{r+1}, \ldots, \mathfrak{m}_{r+s}$ do not contain $\mathfrak{a}$ there exist $x_j \in \mathfrak{a}$ such that $x_j \notin \mathfrak{m}_{r+j}$ $(1 \leq j \leq s)$. Since each $A_{\mathfrak{m}_i}$ $(1 \leq i \leq r)$ is Noetherian, the extension of $\mathfrak{a}$ in $A_{\mathfrak{m}_i}$ is finitely generated. Hence there exist $x_{s+1}, \ldots, x_t$ in $\mathfrak{a}$ whose images in $A_{\mathfrak{m}_i}$ generate $\mathfrak{a}A_{\mathfrak{m}_i}$ for $i = 1, \ldots, r$. Let $\mathfrak{a}_0 = (x_0, \ldots, x_t)$. Show that $\mathfrak{a}_0$ and $\mathfrak{a}$ have the same extension in $A_\mathfrak{m}$ for every maximal ideal $\mathfrak{m}$, and deduce by (3.9) that $\mathfrak{a}_0 = \mathfrak{a}$.]

Solution. Let $\mathfrak{a} \neq 0$ be an ideal (the zero ideal is finitely generated) and choose $\mathfrak{m}_1, \ldots, \mathfrak{m}_{r+s}$, x_0 and $x_1, \ldots, x_s$ as in the hint; by (2) the set of maximal ideals containing x_0 is finite, and it contains those containing $\mathfrak{a}$. For $i \leq r$, $A_{\mathfrak{m}_i}$ is Noetherian by (1), so $\mathfrak{a}A_{\mathfrak{m}_i}$ is generated by finitely many fractions a'/s' with $a' \in \mathfrak{a}$; since s' is a unit, the elements $a'/1$ also generate. Collect all the finitely many elements $a' \in \mathfrak{a}$ so obtained (for all $i \leq r$) as $x_{s+1}, \ldots, x_t$, and let $\mathfrak{a}_0 = (x_0, x_1, \ldots, x_t) \subseteq \mathfrak{a}$, finitely generated.

Let $\mathfrak{m}$ be a maximal ideal; we show $\mathfrak{a}_0 A_\mathfrak{m} = \mathfrak{a}A_\mathfrak{m}$ ("$\subseteq$" is clear).

- $\mathfrak{m} = \mathfrak{m}_i$, $i \leq r$: $\mathfrak{a}_0 A_\mathfrak{m} \supseteq (x_{s+1}, \ldots, x_t)A_\mathfrak{m} = \mathfrak{a}A_\mathfrak{m}$.
- $\mathfrak{m} = \mathfrak{m}_{r+j}$, $1 \leq j \leq s$: $x_j \in \mathfrak{a}_0$ and $x_j \notin \mathfrak{m}$, so $\mathfrak{a}_0 A_\mathfrak{m} = A_\mathfrak{m} \supseteq \mathfrak{a}A_\mathfrak{m}$.
- any other $\mathfrak{m}$: it does not contain x_0, so $x_0 \in \mathfrak{a}_0$ is a unit in $A_\mathfrak{m}$ and $\mathfrak{a}_0 A_\mathfrak{m} = A_\mathfrak{m}$.

Hence $(\mathfrak{a}/\mathfrak{a}_0)_\mathfrak{m} = \mathfrak{a}A_\mathfrak{m}/\mathfrak{a}_0 A_\mathfrak{m} = 0$ for all maximal $\mathfrak{m}$ (3.3), so $\mathfrak{a}/\mathfrak{a}_0 = 0$ by 3.8, i.e. $\mathfrak{a} = \mathfrak{a}_0$ is finitely generated. □

Exercise 10

Problem. Let M be a Noetherian A-module. Show that $M[x]$ (Chapter 2, Exercise 6) is a Noetherian $A[x]$-module.

Solution. Let $N \subseteq M[x]$ be an $A[x]$-submodule. For $n \geq 0$ let $L_n = \{m \in M : m$ is the coefficient of x^n of some N with $\deg f \leq n\}$. L_n is an A-submodule of M, and $L_n \subseteq L_{n+1}$ (use xf). Since M is Noetherian, there is r with $L_n = L_r$ for $n \geq r$, and each L_n $(n \leq r)$ is finitely generated, say by $m_{n,1}, \ldots, m_{n,k_n}$; choose $f_{n,j} \in N$ of degree $\leq n$ with x^n-coefficient $m_{n,j}$. Let $N' \subseteq N$ be the $A[x]$-submodule generated by all $f_{n,j}$ $(n \leq r)$. We show $N = N'$ by induction on $\deg f$ for $f \in N$ (the case $f = 0$ is trivial). Let $\deg f = d$ with leading coefficient $m \in L_d$. If $d \leq r$, write $m = \sum_j a_j m_{d,j}$; then $f - \sum_j a_j f_{d,j} \in N$ has degree $< d$. If $d > r$, $m \in L_d = L_r$, $m = \sum_j a_j m_{r,j}$, and $f - \sum_j a_j x^{d-r} f_{r,j} \in N$ has degree $< d$. By induction the difference lies in N', hence $f \in N'$. So N is finitely generated, and $M[x]$ is Noetherian. □

Exercise 11

Problem. Let A be a ring such that each local ring $A_\mathfrak{p}$ is Noetherian. Is A necessarily Noetherian?

Solution. *No.* Let $A = \prod_{n\geq 1} \mathbf{F}_2$, a Boolean ring. For a prime $\mathfrak{p}$ of A, $A_\mathfrak{p}$ is again Boolean ($s^2 = s$ gives $a/s^2 = a/s$, hence $(a/s)^2 = a^2/s^2 = a/s^2 = a/s$) and local, and a local ring has no idempotent $\neq 0, 1$ (Ch. 1, Ex. 12), so every element of $A_\mathfrak{p}$ is 0 or 1: $A_\mathfrak{p} = \mathbf{F}_2$, Noetherian. But A is not Noetherian: the ideal I of sequences with finite support is not finitely generated, since finitely many elements have supports in a finite set F of coordinates and generate only sequences supported in F, while I contains the unit vectors e_n for $n \notin F$. □

Exercise 12

Problem. Let A be a ring and B a faithfully flat A-algebra (Chapter 3, Exercise 16). If B is Noetherian, show that A is Noetherian. [Use the ascending chain condition.]

Solution. Let $\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \cdots$ be ideals of A. The extensions $\mathfrak{a}_1^e \subseteq \mathfrak{a}_2^e \subseteq \cdots$ form an ascending chain in the Noetherian ring B, so $\mathfrak{a}_n^e = \mathfrak{a}_{n+1}^e = \cdots$ for some n. By faithful flatness $\mathfrak{a}^{ec} = \mathfrak{a}$ for every ideal $\mathfrak{a}$ (Ch. 3, Ex. 16(i)), hence $\mathfrak{a}_m = \mathfrak{a}_m^{ec} = \mathfrak{a}_n^{ec} = \mathfrak{a}_n$ for $m \geq n$. □

Exercise 13

Problem. Let $f : A \to B$ be a ring homomorphism of finite type and let $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the mapping associated with f. Show that the fibers of f^* are Noetherian subspaces of $\mathrm{Spec}(B)$.

Solution. The fibre of f^* over $\mathfrak{p} \in \mathrm{Spec}\, A$ is homeomorphic, as a subspace of $\mathrm{Spec}\, B$, to $\mathrm{Spec}(k(\mathfrak{p}) \otimes_A B)$ (Ch. 3, Ex. 21(iv)). Since B is a finitely generated A-algebra, $k(\mathfrak{p}) \otimes_A B$ is a finitely generated $k(\mathfrak{p})$-algebra, hence a Noetherian ring (7.7), and its spectrum is a Noetherian space. □

Exercise 14

Problem. (*Nullstellensatz, strong form.*) Let k be an algebraically closed field, let A denote the polynomial ring $k[t_1, \ldots, t_n]$ and let $\mathfrak{a}$ be an ideal in A. Let V be the variety in k^n defined by the ideal $\mathfrak{a}$, so that V is the set of all $x = (x_1, \ldots, x_n) \in k^n$ such that $f(x) = 0$ for all $f \in \mathfrak{a}$. Let $I(V)$ be the ideal of V, i.e. the ideal of all polynomials $g \in A$ such that $g(x) = 0$ for all $x \in V$. Then $I(V) = r(\mathfrak{a})$.

[It is clear that $r(\mathfrak{a}) \subseteq I(V)$. Conversely, let $f \notin r(\mathfrak{a})$, then there is a prime ideal $\mathfrak{p}$ containing $\mathfrak{a}$ such that $f \notin \mathfrak{p}$. Let $\bar{f}$ be the image of f in $B = A/\mathfrak{p}$, let $C = B_{\bar{f}} = B[1/\bar{f}]$, and let $\mathfrak{m}$ be a maximal ideal of C. Since C is a finitely generated k-algebra we have $C/\mathfrak{m} \cong k$, by (7.9). The images x_i in $C/\mathfrak{m}$ of the generators t_i of A thus define a point $x = (x_1, \ldots, x_n) \in k^n$, and the construction shows that $x \in V$ and $f(x) \neq 0$.]

Solution. If $f \in r(\mathfrak{a})$, $f^m \in \mathfrak{a}$, then for $x \in V$ we have $f(x)^m = 0$, so $f(x) = 0$; thus $r(\mathfrak{a}) \subseteq I(V)$. Conversely let $f \notin r(\mathfrak{a})$. By 1.14 there is a prime $\mathfrak{p} \supseteq \mathfrak{a}$ with $f \notin \mathfrak{p}$. Let $B = A/\mathfrak{p}$, $\bar{f} \neq 0$ the

image of f, and $C = B[1/\bar{f}]$; it is non-zero because B is a domain and $\bar{f} \neq 0$. Let $\mathfrak{m}$ be a maximal ideal of C. C is a finitely generated k-algebra (generated by the images of the t_i and $1/\bar{f}$), so $C/\mathfrak{m}$ is a finite extension of k (7.10), i.e. $C/\mathfrak{m} = k$. The composite $\varphi : A \to B \to C \to C/\mathfrak{m} = k$ is a k-algebra homomorphism; put $x_i = \varphi(t_i)$ and $x = (x_1, \ldots, x_n)$. Then $\varphi(g) = g(x)$ for every $g \in A$. For $g \in \mathfrak{a} \subseteq \mathfrak{p}$, $\varphi(g) = 0$, so $x \in V$. And $\varphi(f) \neq 0$ because $\bar{f}$ is a unit of C and so maps to a unit of k. Hence $f(x) \neq 0$ and $f \notin I(V)$. So $I(V) \subseteq r(\mathfrak{a})$. □

Exercise 15

Problem. Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal and k its residue field, and let M be a finitely generated A-module. Then the following are equivalent:

(i) M is free;

(ii) M is flat;

(iii) the mapping of $\mathfrak{m} \otimes M$ into $A \otimes M$ is injective;

(iv) $\mathrm{Tor}_1^A(k, M) = 0$.

[To show that iv)⇒i), let $x_1, \ldots, x_n$ be elements of M whose images in $M/\mathfrak{m}M$ form a k-basis of this vector space. By (2.8), the x_i generate M. Let F be a free A-module with basis $e_1, \ldots, e_n$ and define $\phi : F \to M$ by $\phi(e_i) = x_i$. Let $E = \mathrm{Ker}(\phi)$. Then the exact sequence $0 \to E \to F \to M \to 0$ gives us an exact sequence $0 \to k \otimes_A E \to k \otimes_A F \xrightarrow{1 \otimes \phi} k \otimes_A M \to 0$. Since $k \otimes F$ and $k \otimes M$ are vector spaces of the same dimension over k, it follows that $1 \otimes \phi$ is an isomorphism, hence $k \otimes E = 0$, hence $E = 0$ by Nakayama's Lemma (E is finitely generated because it is a submodule of F, and A is Noetherian).]

Solution. (i)⇒(ii): a free module is a direct sum of copies of the flat module A, hence flat (Ch. 2, Ex. 4).

(ii)⇒(iii): tensor the injection $\mathfrak{m} \to A$ with the flat module M.

(iii)⇒(iv): from $0 \to \mathfrak{m} \to A \to k \to 0$ we get the exact sequence $\mathrm{Tor}_1(A, M) \to \mathrm{Tor}_1(k, M) \to \mathfrak{m} \otimes M \to A \otimes M$, and $\mathrm{Tor}_1(A, M) = 0$. So $\mathrm{Tor}_1(k, M) = \mathrm{Ker}(\mathfrak{m} \otimes M \to A \otimes M) = 0$.

(iv)⇒(i): Since M is finitely generated, $M/\mathfrak{m}M = k \otimes M$ is a finite-dimensional k-vector space; choose $x_1, \ldots, x_n \in M$ whose images form a basis. By Nakayama (2.8) they generate M. Let $\phi : F = A^n \to M$, $e_i \mapsto x_i$, surjective, with kernel E. The Tor sequence of $0 \to E \to F \to M \to 0$ gives the exact sequence $\mathrm{Tor}_1(k, M) \to k \otimes E \to k \otimes F \to k \otimes M \to 0$, and $\mathrm{Tor}_1(k, M) = 0$ by (iv). Now $1 \otimes \phi : k^n \to M/\mathfrak{m}M$ sends a basis to a basis, so it is an isomorphism; by exactness $k \otimes E = 0$, i.e. $E = \mathfrak{m}E$. E is finitely generated (a submodule of F over the Noetherian ring A), so $E = 0$ by Nakayama. Hence ϕ is an isomorphism and M is free. □

Exercise 16

Problem. Let A be a Noetherian ring, M a finitely generated A-module. Then the following are equivalent:

(i) M is a flat A-module;

(ii) $M_\mathfrak{p}$ is a free $A_\mathfrak{p}$-module, for all prime ideals $\mathfrak{p}$;

(iii) $M_{\mathfrak{m}}$ is a free $A_{\mathfrak{m}}$-module, for all maximal ideals $\mathfrak{m}$.

In other words, flat = locally free. [Use Exercise 15.]

Solution. (i)⇒(ii): $M_{\mathfrak{p}}$ is flat over $A_{\mathfrak{p}}$ (3.10), finitely generated over the Noetherian local ring $A_{\mathfrak{p}}$ (7.4), hence free by Ex. 15. (ii)⇒(iii): maximal ideals are prime. (iii)⇒(i): a free $A_{\mathfrak{m}}$-module is flat, so M is flat by 3.10(iii). □

Exercise 17

Problem. Let A be a ring and M a Noetherian A-module. Show (by imitating the proofs of (7.11) and (7.12)) that every submodule N of M has a primary decomposition (Chapter 4, Exercises 20–23).

Solution. Call a proper submodule $Z \subsetneq M$ *irreducible* if $Z = X \cap Y$ (X, Y submodules) implies $Z = X$ or $Z = Y$.

Step 1 (analogue of 7.11): every proper submodule is a finite intersection of irreducible submodules. If not, the set of proper submodules for which this fails is non-empty and has a maximal element Z (M Noetherian). Z is not irreducible, so $Z = X \cap Y$ with $X, Y \supsetneq Z$ (and proper, since $X = M$ would give $Z = Y$); by maximality X, Y are finite intersections of irreducibles, hence so is Z, a contradiction.

Step 2 (analogue of 7.12): an irreducible submodule Z is primary. Put $W = M/Z \neq 0$, a Noetherian module in which 0 is irreducible. Let $x \in A$ be a zero-divisor in W: $xw = 0$ with $w \neq 0$. The chain of submodules $\operatorname{Ker}(x^n : W \to W)$ is ascending, so $\operatorname{Ker} x^n = \operatorname{Ker} x^{n+1}$ for some n. Then $x^n W \cap Aw = 0$: if $y = x^n w' = aw$, then $xy = axw = 0$, so $x^{n+1}w' = 0$, so $x^n w' = 0$, i.e. $y = 0$. Since 0 is irreducible and $Aw \neq 0$, we conclude $x^n W = 0$: x is nilpotent in W. So Z is primary in M.

Hence for a submodule $N \neq M$, apply Steps 1–2 to the Noetherian module M/N and the proper submodule 0: $0 = \bigcap \bar{Z}_i$ with $\bar{Z}_i$ irreducible, hence primary in M/N; their preimages $Z_i \supseteq N$ give $N = \bigcap Z_i$ with Z_i primary in M (since $M/Z_i \cong (M/N)/\bar{Z}_i$). If $N = M$ the empty intersection works. □

Exercise 18

Problem. Let A be a Noetherian ring, $\mathfrak{p}$ a prime ideal of A, and M a finitely generated A-module. Show that the following are equivalent:

(i) $\mathfrak{p}$ belongs to 0 in M;

(ii) there exists $x \in M$ such that $\operatorname{Ann}(x) = \mathfrak{p}$;

(iii) there exists a submodule of M isomorphic to $A/\mathfrak{p}$.

Deduce that there exists a chain of submodules $0 = M_0 \subset M_1 \subset \cdots \subset M_r = M$ such that each quotient M_i/M_{i-1} is of the form $A/\mathfrak{p}_i$, where $\mathfrak{p}_i$ is a prime ideal of A.

Solution. By Ex. 17 0 has a primary decomposition in the Noetherian module M; by the analogue of 4.5 (Ch. 4, Ex. 22) the primes belonging to 0 in M are exactly those among the ideals $r(\operatorname{Ann} x)$, $x \in M$, which are prime.

(ii)$\Leftrightarrow$(iii): $Ax \cong A/\operatorname{Ann}(x)$; conversely if $\varphi : A/\mathfrak{p} \to M$ is an injective homomorphism, then $x = \varphi(1)$ has $\operatorname{Ann} x = \mathfrak{p}$.

(ii)$\Rightarrow$(i): $x \neq 0$ ($\mathfrak{p} \neq A$) and $r(\operatorname{Ann} x) = \mathfrak{p}$ is prime, hence belongs to 0.

(i)$\Rightarrow$(ii) (imitating 7.17): let $0 = Q_1 \cap \cdots \cap Q_n$ be a minimal primary decomposition with $r_M(Q_i) = \mathfrak{p}_i$ and $\mathfrak{p} = \mathfrak{p}_1$. By minimality $N_1 = Q_2 \cap \cdots \cap Q_n \neq 0$ and $N_1 \cap Q_1 = 0$. For $0 \neq x \in N_1$: $x \in Q_j$ ($j \geq 2$) so $(Q_j : x) = (1)$, and $x \notin Q_1$; hence $\operatorname{Ann}(x) = \bigcap_j (Q_j : x) = (Q_1 : x)$, which is $\mathfrak{p}_1$-primary (Ch. 4, Ex. 21), so $\operatorname{Ann}(x) \subseteq \mathfrak{p}_1$. Since $\mathfrak{p}_1 = r(Q_1 : M)$ and A is Noetherian, $\mathfrak{p}_1^m \subseteq (Q_1 : M)$ for some m (7.14), so $\mathfrak{p}_1^m N_1 \subseteq Q_1 \cap N_1 = 0$. Let $m \geq 1$ be minimal with $\mathfrak{p}_1^m N_1 = 0$ and pick $0 \neq x \in \mathfrak{p}_1^{m-1} N_1$ (where $\mathfrak{p}_1^0 = A$). Then $\mathfrak{p}_1 x \subseteq \mathfrak{p}_1^m N_1 = 0$, so $\mathfrak{p}_1 \subseteq \operatorname{Ann}(x) \subseteq \mathfrak{p}_1$, i.e. $\operatorname{Ann}(x) = \mathfrak{p}_1$.

The chain. If $M \neq 0$, then $0 \neq M$ has a primary decomposition with at least one component, so some prime belongs to 0, and by (iii) M has a submodule $M_1 \cong A/\mathfrak{p}_1$. Apply the same to the finitely generated module M/M_1 (if non-zero) to get $M_2 \supset M_1$ with $M_2/M_1 \cong A/\mathfrak{p}_2$, and so on. The ascending chain $M_1 \subsetneq M_2 \subsetneq \cdots$ must stop since M is Noetherian, which happens only when $M_r = M$. □

Exercise 19

> **Problem.** Let $\mathfrak{a}$ be an ideal in a Noetherian ring A. Let $\mathfrak{a} = \bigcap_{i=1}^{r} \mathfrak{b}_i = \bigcap_{j=1}^{s} \mathfrak{c}_j$ be two minimal decompositions of $\mathfrak{a}$ as intersections of *irreducible* ideals. Prove that $r = s$ and that (possibly after re-indexing the $\mathfrak{c}_j$) $r(\mathfrak{b}_i) = r(\mathfrak{c}_i)$ for all i. [Show that for each $i = 1, \ldots, r$ there exists j such that $\mathfrak{a} = \mathfrak{b}_1 \cap \cdots \cap \mathfrak{b}_{i-1} \cap \mathfrak{c}_j \cap \mathfrak{b}_{i+1} \cap \cdots \cap \mathfrak{b}_r$.]
>
> State and prove an analogous result for modules.

Solution. I prove the module version, which contains the ideal version (take $M = A$ and $N = \mathfrak{a}$; the submodules of A are the ideals, irreducible ideals are irreducible submodules, and $r_A(\mathfrak{b}) = r(\mathfrak{b})$). The argument does not use the replacement property suggested in the hint; instead it computes the number of components with a given radical.

Statement. Let A be a Noetherian ring, M a finitely generated A-module, N a submodule, and let $N = Z_1 \cap \cdots \cap Z_r = Z_1' \cap \cdots \cap Z_s'$ be two minimal (irredundant) decompositions into irreducible submodules. Then $r = s$ and, after re-indexing, $r_M(Z_i) = r_M(Z_i')$ for all i.

Proof. Replacing M by M/N and Z_i by Z_i/N we may assume $N = 0$ (this preserves irreducibility, irredundancy and r_M). So let $P = M$ and $0 = Z_1 \cap \cdots \cap Z_r$ be irredundant. By Ex. 17, each Z_i is primary; put $\mathfrak{p}_i = r_P(Z_i) = r(\operatorname{Ann}(P/Z_i))$, a prime. Let $W_i = P/Z_i$ and $N_i = \bigcap_{k \neq i} Z_k$; irredundancy says $N_i \neq 0$, and $N_i \cap Z_i = 0$.

For a prime $\mathfrak{q}$ let $k(\mathfrak{q}) = A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q}$ and for an A-module U let $\operatorname{Soc}_\mathfrak{q}(U) = \{z \in U_\mathfrak{q} : \mathfrak{q}A_\mathfrak{q}\, z = 0\}$, a $k(\mathfrak{q})$-vector space. I show

$$\dim_{k(\mathfrak{p})} \operatorname{Soc}_\mathfrak{p}(P) = \#\{i : \mathfrak{p}_i = \mathfrak{p}\} \qquad \text{for every prime } \mathfrak{p}. \tag{$*$}$$

The right side is then determined by P alone, so it is the same for both decompositions, which gives $r = s$ and the equality of radicals after re-indexing.

(a) Facts about one $W = W_i$ with $\mathfrak{p}_i = \mathfrak{p}_0$. 0 is an irreducible primary submodule of W, so any two non-zero submodules of W meet non-trivially, and for $0 \neq w \in W$ we have $r(\operatorname{Ann} w) = \mathfrak{p}_0$ (Ch. 4, Ex. 21, analogue of 4.4(ii)). Claim: $\operatorname{Soc}_{\mathfrak{q}}(W) = 0$ if $\mathfrak{q} \neq \mathfrak{p}_0$, and $\dim_{k(\mathfrak{p}_0)} \operatorname{Soc}_{\mathfrak{p}_0}(W) \leq 1$. Let $0 \neq z \in \operatorname{Soc}_{\mathfrak{q}}(W)$, $z = w/s$. If $\mathfrak{q} = (u_1, \ldots, u_l)$ then $u_k z = 0$ gives $t_k \notin \mathfrak{q}$ with $t_k u_k w = 0$; with $t = \prod t_k$, $\mathfrak{q} \cdot (tw) = 0$. Also $tw/1 = t \cdot sz \neq 0$ (a non-zero element times units), so $w' := tw$ satisfies $\mathfrak{q} w' = 0$ and $w'/1 \neq 0$. Then $\mathfrak{q} \subseteq \operatorname{Ann}(w')$, while $\operatorname{Ann}(w') \subseteq \mathfrak{q}$ (if $b \in \operatorname{Ann} w'$ with $b \notin \mathfrak{q}$, then $w'/1 = 0$). So $\operatorname{Ann}(w') = \mathfrak{q}$, and $\mathfrak{q} = r(\operatorname{Ann} w') = \mathfrak{p}_0$. This proves the first assertion, and shows that $\operatorname{Soc}_{\mathfrak{p}_0}(W)$ is spanned over $k(\mathfrak{p}_0)$ by classes $w/1$ with $w \in W$, $\mathfrak{p}_0 w = 0$, $w \neq 0$ (so $\operatorname{Ann} w = \mathfrak{p}_0$). For two such w, w' we have $Aw \cap Aw' \neq 0$, i.e. $aw = a'w' \neq 0$; as $\mathfrak{p}_0 w = 0$, $a, a' \notin \mathfrak{p}_0$ are units in $A_{\mathfrak{p}_0}$, so $w/1$ and $w'/1$ are proportional over $A_{\mathfrak{p}_0}$. Hence the dimension is ≤ 1.

(b) Upper bound. $0 = \bigcap Z_i$ gives an injection $P \to \bigoplus W_i$, hence (localization is exact) $P_{\mathfrak{p}} \hookrightarrow \bigoplus (W_i)_{\mathfrak{p}}$ and $\operatorname{Soc}_{\mathfrak{p}}(P) \hookrightarrow \bigoplus_i \operatorname{Soc}_{\mathfrak{p}}(W_i)$. By (a) the i-th summand is 0 unless $\mathfrak{p}_i = \mathfrak{p}$ and has dimension ≤ 1 otherwise. So $\dim \operatorname{Soc}_{\mathfrak{p}}(P) \leq \#\{i : \mathfrak{p}_i = \mathfrak{p}\}$.

(c) Lower bound. Fix i. Since $N_i \cap Z_i = 0$, N_i embeds in W_i; as A is Noetherian, $\mathfrak{p}_i^m \subseteq (Z_i : P) = \operatorname{Ann}(W_i)$ for some m (7.14), so $\mathfrak{p}_i^m N_i = 0$. Taking $m \geq 1$ minimal and $0 \neq y_i \in \mathfrak{p}_i^{m-1} N_i$ we get $\mathfrak{p}_i y_i = 0$ and $\operatorname{Ann}(y_i) \subseteq \mathfrak{p}_i$ (as in Ex. 18), so $\operatorname{Ann} y_i = \mathfrak{p}_i$. Let $I_{\mathfrak{p}} = \{i : \mathfrak{p}_i = \mathfrak{p}\}$. Then $\mathfrak{p} y_i = 0$ for $i \in I_{\mathfrak{p}}$, so $y_i/1 \in \operatorname{Soc}_{\mathfrak{p}}(P)$, and it is non-zero: $y_i/1 = 0$ would give $s \notin \mathfrak{p}$ with $s y_i = 0$, i.e. $s \in \operatorname{Ann} y_i = \mathfrak{p}$. We show these are $k(\mathfrak{p})$-independent. Suppose $\sum_{i \in I_{\mathfrak{p}}} c_i \, y_i/1 = 0$ with $c_i = a_i/s$. Then there is $t \notin \mathfrak{p}$ with $t \sum a_i y_i = 0$ in P. Fix $k \in I_{\mathfrak{p}}$ and project to $W_k = P/Z_k$: for $i \neq k$, $y_i \in N_i \subseteq Z_k$ maps to 0, so $t a_k \bar{y}_k = 0$ in W_k, with $\bar{y}_k \neq 0$ (as $N_k \hookrightarrow W_k$). Thus $t a_k \in \operatorname{Ann}(\bar{y}_k) = (Z_k : y_k)$, which is $\mathfrak{p}$-primary (Ch. 4, Ex. 21); since $t \notin \mathfrak{p}$, $a_k \in (Z_k : y_k) \subseteq \mathfrak{p}$, i.e. $c_k \in \mathfrak{p} A_{\mathfrak{p}}$. So all $c_k = 0$ in $k(\mathfrak{p})$, and $\dim \operatorname{Soc}_{\mathfrak{p}}(P) \geq \# I_{\mathfrak{p}}$.

(b) and (c) prove $(*)$. □

Remark. The replacement property from the hint is also true: if $\mathfrak{a}_i' = \bigcap_{k \neq i} \mathfrak{b}_k$, there is j with $\mathfrak{a}_i' \cap \mathfrak{c}_j = \mathfrak{a}$. (If not, each $\mathfrak{a}_i' \cap \mathfrak{c}_j \supsetneq \mathfrak{a}$ is not contained in $\mathfrak{b}_i$; for two ideals $\mathfrak{e}, \mathfrak{e}'$ with $\mathfrak{a} \subseteq \mathfrak{e}, \mathfrak{e}' \subseteq \mathfrak{a}_i'$ not contained in $\mathfrak{b}_i$, irreducibility of $\mathfrak{b}_i$ gives $z \in (\mathfrak{e} + \mathfrak{b}_i) \cap (\mathfrak{e}' + \mathfrak{b}_i) \setminus \mathfrak{b}_i$, $z = u + \beta = u' + \beta'$ with $u \in \mathfrak{e}$, $u' \in \mathfrak{e}'$, $\beta, \beta' \in \mathfrak{b}_i$, and $u - u' \in \mathfrak{a}_i' \cap \mathfrak{b}_i = \mathfrak{a} \subseteq \mathfrak{e}'$ shows $u \in \mathfrak{e} \cap \mathfrak{e}' \setminus \mathfrak{b}_i$; by induction on j some element of $\bigcap_j (\mathfrak{a}_i' \cap \mathfrak{c}_j)$ lies outside $\mathfrak{b}_i$, whereas $\bigcap_j (\mathfrak{a}_i' \cap \mathfrak{c}_j) = \mathfrak{a} \subseteq \mathfrak{b}_i$, absurd.) It does not by itself give the count $r = s$ without further work, which is why I used $(*)$.

Exercise 20

Problem. Let X be a topological space and let $\mathcal{F}$ be the smallest collection of subsets of X which contains all open subsets of X and is closed with respect to the formation of finite intersections and complements.

(i) Show that a subset E of X belongs to $\mathcal{F}$ if and only if E is a finite union of sets of the form $U \cap C$, where U is open and C is closed.

(ii) Suppose that X is irreducible and let $E \in \mathcal{F}$. Show that E is dense in X (that is, that $\bar{E} = X$) if and only if E contains a non-empty open set in X.

Solution. (i) Let $\mathcal{F}'$ be the collection of finite unions of sets $U \cap C$. It contains all open sets ($C = X$) and all closed sets. It is closed under finite intersections: $\big(\bigcup_i U_i \cap C_i\big) \cap \big(\bigcup_j U_j' \cap C_j'\big) = \bigcup_{i,j} (U_i \cap U_j') \cap (C_i \cap C_j')$. It is closed under complements: $X \setminus (U \cap C) = (X \setminus U) \cup (X \setminus C)$ is the

union of a closed set and an open set, so lies in $\mathcal{F}'$, and the complement of a finite union is the finite intersection of the complements. By minimality $\mathcal{F} \subseteq \mathcal{F}'$. Conversely $\mathcal{F}$ contains open sets, hence (complements) closed sets, hence $U \cap C$ (intersections), and finite unions (complement of the intersection of complements). So $\mathcal{F}' \subseteq \mathcal{F}$.

(ii) Write $E = \bigcup_{i=1}^{n}(U_i \cap C_i)$ with every $U_i \cap C_i \neq \emptyset$ (possible since $E \neq \emptyset$ when E is dense in $X \neq \emptyset$). ($\Leftarrow$) A non-empty open set in an irreducible space is dense, so $\bar{E} \supseteq \overline{U} = X$. ($\Rightarrow$) If $\bar{E} = X$, then $X = \bigcup_i \overline{U_i \cap C_i}$, a finite union of closed sets; X is irreducible, so $X = \overline{U_i \cap C_i} \subseteq C_i$ for some i. Then $C_i = X$ and $U_i = U_i \cap C_i \subseteq E$ is a non-empty open subset of E. □

Exercise 21

> **Problem.** Let X be a Noetherian topological space (Chapter 6, Exercise 5) and let $E \subseteq X$. Show that $E \in \mathcal{F}$ if and only if, for each irreducible closed set $X_0 \subseteq X$, either $\overline{E \cap X_0} \neq X_0$ or else $E \cap X_0$ contains a non-empty open subset of X_0. [Suppose $E \notin \mathcal{F}$. Then the collection of closed sets $X' \subseteq X$ such that $E \cap X' \notin \mathcal{F}$ is not empty and therefore has a minimal element X_0. Show that X_0 is irreducible and then that each of the alternatives above leads to the conclusion that $E \cap X_0 \in \mathcal{F}$.] The sets belonging to $\mathcal{F}$ are called the *constructible* subsets of X.

Solution. ($\Rightarrow$) Let $E \in \mathcal{F}$ and X_0 irreducible closed. By Ex. 20(i), $E = \bigcup(U_i \cap C_i)$, so $E \cap X_0 = \bigcup(U_i \cap X_0) \cap (C_i \cap X_0)$ is a finite union of sets (open in X_0)∩(closed in X_0), i.e. belongs to the family $\mathcal{F}(X_0)$ of constructible subsets of the space X_0, which is irreducible. If $\overline{E \cap X_0} = X_0$, then by Ex. 20(ii) $E \cap X_0$ contains a non-empty open subset of X_0.

($\Leftarrow$) Suppose the condition holds but $E \notin \mathcal{F}$. The collection of closed X' with $E \cap X' \notin \mathcal{F}$ contains X; as X is Noetherian it has a minimal element X_0. $X_0 \neq \emptyset$ since $E \cap \emptyset = \emptyset \in \mathcal{F}$. X_0 is irreducible: if $X_0 = X_1 \cup X_2$ with $X_i \subsetneq X_0$ closed, then by minimality $E \cap X_i \in \mathcal{F}$, so $E \cap X_0 \in \mathcal{F}$ (finite union), a contradiction. By hypothesis there are two cases.

- $X_0' := \overline{E \cap X_0} \neq X_0$. Then $X_0' \subsetneq X_0$ is closed, so $E \cap X_0' \in \mathcal{F}$ by minimality; but $E \cap X_0 \subseteq X_0'$, so $E \cap X_0' = E \cap X_0 \in \mathcal{F}$, a contradiction.
- $E \cap X_0$ contains a non-empty set $X_0 \cap U$ (U open in X). Let $X_0'' = X_0 \setminus U$, closed and $\neq X_0$. Then $E \cap X_0 = (X_0 \cap U) \cup (E \cap X_0'')$, where $X_0 \cap U \in \mathcal{F}$ and $E \cap X_0'' \in \mathcal{F}$ by minimality; so $E \cap X_0 \in \mathcal{F}$, a contradiction.

Hence $E \in \mathcal{F}$. □

Exercise 22

> **Problem.** Let X be a Noetherian topological space and let E be a subset of X. Show that E is open in X if and only if, for each irreducible closed subset X_0 in X, either $E \cap X_0 = \emptyset$ or else $E \cap X_0$ contains a non-empty open subset of X_0. [The proof is similar to that of Exercise 21.]

Solution. ($\Rightarrow$) If E is open, $E \cap X_0$ is open in X_0: it is empty or is itself a non-empty open subset of X_0.

($\Leftarrow$) Suppose the condition holds and E is not open. The collection of closed $X' \subseteq X$ such that $E \cap X'$ is not open in X' contains X, so has a minimal element X_0 (X Noetherian). X_0 is irreducible: if $X_0 = X_1 \cup X_2$ with closed proper subsets X_i, then $E \cap X_i$ is open in X_i by minimality, so $X_i \setminus E$ is closed in X_i, hence in X; then $X_0 \setminus E = \bigcup (X_i \setminus E)$ is closed, i.e. $E \cap X_0$ is open in X_0, a contradiction. By hypothesis, $E \cap X_0 \neq \emptyset$ (the empty set is open) and it contains a non-empty open subset $V = X_0 \cap U$ of X_0 (U open in X). Let $X_0' = X_0 \setminus V$, closed and $\neq X_0$. By minimality $E \cap X_0' = X_0' \cap G$ for some open $G \subseteq X$. Then

$$E \cap X_0 = V \cup (E \cap X_0') = V \cup (X_0' \cap G) = V \cup (X_0 \cap G)$$

(indeed $X_0 \cap G = (V \cap G) \cup (X_0' \cap G)$, and $V \cap G \subseteq V$), which is open in X_0. This is a contradiction. Hence E is open. $\square$

Exercise 23

> **Problem.** Let A be a Noetherian ring, $f : A \to B$ a ring homomorphism of finite type (so that B is Noetherian). Let $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$ and let $f^* : Y \to X$ be the mapping associated with f. Then the image under f^* of a constructible subset E of Y is a constructible subset of X.
>
> [By Exercise 20 it is enough to take $E = U \cap C$ where U is open and C is closed in Y; then, replacing B by a homomorphic image, we reduce to the case where E is open in Y. Since Y is Noetherian, E is quasi-compact and therefore a finite union of open sets of the form $\mathrm{Spec}(B_g)$. Hence reduce to the case $E = Y$. To show that $f^*(Y)$ is constructible, use the criterion of Exercise 21. Let X_0 be an irreducible closed subset of X such that $f^*(Y) \cap X_0$ is dense in X_0. We have $f^*(Y) \cap X_0 = f^*(f^{*-1}(X_0))$, and $f^{*-1}(X_0) = \mathrm{Spec}((A/\mathfrak{p}) \otimes_A B)$, where $X_0 = \mathrm{Spec}(A/\mathfrak{p})$. Hence reduce to the case where A is an integral domain and f is injective. If $Y_1, \ldots, Y_n$ are the irreducible components of Y, it is enough to show that some $f^*(Y_i)$ contains a non-empty open set in X. So finally we are brought down to the situation in which A, B are integral domains and f is injective (and still of finite type); now use Chapter 5, Exercise 21 to complete the proof.]

Solution. *Reductions.* By Ex. 20(i) every $E \in \mathcal{F}(Y)$ is a finite union of sets $U \cap C$, and f^* commutes with unions while $\mathcal{F}(X)$ is closed under finite unions; so let $E = U \cap C$. Write $C = V(\mathfrak{b}) \cong \mathrm{Spec}(B/\mathfrak{b})$ (Ch. 1, Ex. 21(iv)); $A \to B/\mathfrak{b}$ is of finite type, and under this homeomorphism $f^*|_C$ is the map induced by $A \to B/\mathfrak{b}$ (Ch. 3, Ex. 21(iii)) and E is open in C. So we may assume E open in Y. Y is Noetherian, so E is quasi-compact; as the sets $Y_g = \mathrm{Spec}(B_g)$ form a basis (Ch. 1, Ex. 17), $E = Y_{g_1} \cup \cdots \cup Y_{g_m}$ and $f^*(E) = \bigcup f^*(Y_{g_j})$, the image of $\mathrm{Spec}(B_{g_j}) \to X$; $A \to B_{g_j}$ is again of finite type. So it suffices to show: *for f of finite type, $E = f^*(Y) \in \mathcal{F}(X)$.*

Criterion. We use Ex. 21 (X is Noetherian as A is). Let X_0 be an irreducible closed subset of X; then $X_0 = V(\mathfrak{p})$ with $\mathfrak{p}$ prime (Ch. 1, Ex. 20(iv)), $X_0 \cong \mathrm{Spec}(A/\mathfrak{p})$. We have $E \cap X_0 = f^*(f^{*-1}(X_0))$ with $f^{*-1}(X_0) = V(\mathfrak{p}B) \cong \mathrm{Spec}(B/\mathfrak{p}B)$, and $E \cap X_0$ is the image of the map induced by $\bar{f} : A/\mathfrak{p} \to B/\mathfrak{p}B$ (Ch. 3, Ex. 21(iii)), again of finite type. So we may replace A by the Noetherian domain $A/\mathfrak{p}$, B by $B/\mathfrak{p}B$, X_0 by X, and must show: *if A is a domain, then either $\overline{f^*(Y)} \neq X$ or $f^*(Y)$ contains a non-empty open set of X.* By Ch. 1, Ex. 21(iii), $\overline{f^*(Y)} = V(\mathrm{Ker}\, f)$. If $\mathrm{Ker}\, f \neq 0$, then $(0) \notin V(\mathrm{Ker}\, f)$, so the closure is not X: the first alternative. If $\mathrm{Ker}\, f = 0$: B is Noetherian with minimal primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_n$ (the irreducible components $Y_i = V(\mathfrak{q}_i)$ of Y) and $\bigcap \mathfrak{q}_i = \mathfrak{N}(B)$, so $\bigcap_i f^{-1}(\mathfrak{q}_i) = f^{-1}(\mathfrak{N}B) = \mathfrak{N}(A) = 0$.

As A is a domain and $\prod f^{-1}(\mathfrak{q}_i) \subseteq \bigcap f^{-1}(\mathfrak{q}_i) = 0$, some $f^{-1}(\mathfrak{q}_i) = 0$. Then $A \hookrightarrow B/\mathfrak{q}_i$ is an injection of domains of finite type, and $f^*(Y_i)$ is the image of $\mathrm{Spec}(B/\mathfrak{q}_i) \to X$. By the key lemma Ch. 5, Ex. 21 there is $u \neq 0$ in A such that for every prime $\mathfrak{p}' \not\ni u$ the map $A \to A/\mathfrak{p}' \to \Omega = \overline{\mathrm{Frac}(A/\mathfrak{p}')}$ extends to $g : B/\mathfrak{q}_i \to \Omega$; then $\mathfrak{q}' = \mathrm{Ker}\, g$ is a prime of $B/\mathfrak{q}_i$ with $\mathfrak{q}'^c = \mathfrak{p}'$. So $X_u \subseteq f^*(Y_i) \subseteq f^*(Y)$, a non-empty open set (as $u \neq 0$ and A is a domain, $(0) \in X_u$): the second alternative. By Ex. 21, $f^*(Y) \in \mathcal{F}(X)$. □

Exercise 24

> **Problem.** With the notation and hypotheses of Exercise 23, f^* is an open mapping $\iff$ f has the going-down property (Chapter 5, Exercise 10). [Suppose f has the going-down property. As in Exercise 23, reduce to proving that $E = f^*(Y)$ is open in X. The going-down property asserts that if $\mathfrak{p} \in E$ and $\mathfrak{p}' \subseteq \mathfrak{p}$, then $\mathfrak{p}' \in E$: in other words, that if X_0 is an irreducible closed subset of X and X_0 meets E, then $E \cap X_0$ is dense in X_0. By Exercises 20 and 22, E is open in X.]

Solution. Recall the going-down property: if $\mathfrak{p}' \subseteq \mathfrak{p}$ are primes of A and $\mathfrak{q}$ is a prime of B with $\mathfrak{q}^c = \mathfrak{p}$, then there is a prime $\mathfrak{q}' \subseteq \mathfrak{q}$ of B with $\mathfrak{q}'^c = \mathfrak{p}'$.

($\Rightarrow$) Let f^* be open, $\mathfrak{p}' \subseteq \mathfrak{p}$, $\mathfrak{q}^c = \mathfrak{p}$. For $g \in B \setminus \mathfrak{q}$ the set Y_g is an open neighbourhood of $\mathfrak{q}$, so $f^*(Y_g)$ is an open set containing $\mathfrak{p}$. Since $\mathfrak{p} \in V(\mathfrak{p}') = \overline{\{\mathfrak{p}'\}}$ (Ch. 1, Ex. 18), every open set containing $\mathfrak{p}$ contains $\mathfrak{p}'$; hence $T \cap Y_g \neq \emptyset$ for all $g \notin \mathfrak{q}$, where $T = f^{*-1}(\mathfrak{p}')$ is the fibre over $\mathfrak{p}'$. $T \cong \mathrm{Spec}(R)$ with $R = k(\mathfrak{p}') \otimes_A B$ Noetherian (Ex. 13, Ch. 3, Ex. 21(iv)), so R has finitely many minimal primes and every prime of R contains one. Since for $x, y \in T$ we have $y \in \overline{\{x\}}$ iff $\mathfrak{q}_x \subseteq \mathfrak{q}_y$ (Ch. 1, Ex. 18(iii)), a condition independent of the ambient space, there are finitely many $\mathfrak{q}_1, \ldots, \mathfrak{q}_r \in T$ such that every element of T contains some $\mathfrak{q}_j$. Suppose no $\mathfrak{q}_j$ is contained in $\mathfrak{q}$. Choose $g_j \in \mathfrak{q}_j \setminus \mathfrak{q}$ and put $g = g_1 \cdots g_r$; then $g \notin \mathfrak{q}$ since $\mathfrak{q}$ is prime. Every $\mathfrak{z} \in T$ contains some $\mathfrak{q}_j \ni g$, so $\mathfrak{z} \notin Y_g$, i.e. $T \cap Y_g = \emptyset$, contradicting the above. Hence some $\mathfrak{q}_j \subseteq \mathfrak{q}$, and $\mathfrak{q}_j \in T$ means $\mathfrak{q}_j^c = \mathfrak{p}'$. So f has going-down.

($\Leftarrow$) Suppose f has going-down. Every open set of Y is a union of sets $Y_g \cong \mathrm{Spec}(B_g)$, so it suffices to show $f^*(Y_g)$ is open for each g. The map $f_g : A \to B_g$ is of finite type, and it has going-down: if $\mathfrak{p}' \subseteq \mathfrak{p}$ and $\mathfrak{q} \in Y_g$ lies over $\mathfrak{p}$, going-down for f gives $\mathfrak{q}' \subseteq \mathfrak{q}$ over $\mathfrak{p}'$, and $g \notin \mathfrak{q} \supseteq \mathfrak{q}'$ so $\mathfrak{q}' \in Y_g$. So it suffices to show: *if f is of finite type with going-down, then $E = f^*(Y)$ is open.* By Ex. 23, E is constructible. Let X_0 be an irreducible closed subset of X meeting E; $X_0 = V(\mathfrak{p}_0)$ with $\mathfrak{p}_0$ prime. If $\mathfrak{p} \in E \cap X_0$, then $\mathfrak{p} \supseteq \mathfrak{p}_0$ and going-down gives $\mathfrak{p}_0 \in E$. So $E \cap X_0$ contains the generic point $\mathfrak{p}_0$ of X_0 and is therefore dense in X_0. As $E \cap X_0$ is constructible in X_0 (proof of Ex. 21) and X_0 is irreducible, Ex. 20(ii) shows that $E \cap X_0$ contains a non-empty open subset of X_0. By Ex. 22 (every irreducible closed X_0 either misses E or satisfies this), E is open in X. □

Exercise 25

> **Problem.** Let A be Noetherian, $f : A \to B$ of finite type and flat (i.e., B is flat as an A-module). Then $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is an open mapping. [Exercise 24 and Chapter 5, Exercise 11.]

Solution. By Ex. 24 it suffices to show that f has the going-down property. Let $\mathfrak{p}' \subseteq \mathfrak{p}$ be primes of A and $\mathfrak{q}$ a prime of B with $\mathfrak{q}^c = \mathfrak{p}$. Since f is flat, $f^* : \mathrm{Spec}(B_\mathfrak{q}) \to \mathrm{Spec}(A_\mathfrak{p})$ is surjective (Ch. 3, Ex. 18). The prime $\mathfrak{p}'A_\mathfrak{p}$ of $A_\mathfrak{p}$ therefore is the contraction of a prime Q of $B_\mathfrak{q}$; $Q = \mathfrak{q}'B_\mathfrak{q}$ for a prime $\mathfrak{q}' \subseteq \mathfrak{q}$ of B (3.13). The composites $A \to A_\mathfrak{p} \to B_\mathfrak{q}$ and $A \to B \to B_\mathfrak{q}$ coincide, so the contraction of Q to A is both $(\mathfrak{p}'A_\mathfrak{p})^c = \mathfrak{p}'$ and $\mathfrak{q}'^c$. Hence $\mathfrak{q}'^c = \mathfrak{p}'$, and f has going-down. (This is exactly the content of Ch. 5, Ex. 11 that flat homomorphisms have the going-down property.) □

Exercise 26

Problem. (*Grothendieck groups.*) Let A be a Noetherian ring and let $F(A)$ denote the set of all isomorphism classes of finitely generated A-modules. Let C be the free abelian group generated by $F(A)$. With each short exact sequence $0 \to M' \to M \to M'' \to 0$ of finitely generated A-modules we associate the element $(M') - (M) + (M'')$ of C, where (M) is the isomorphism class of M, etc. Let D be the subgroup of C generated by these elements, for all short exact sequences. The quotient group C/D is called the *Grothendieck group* of A, and is denoted by $K(A)$. If M is a finitely generated A-module, let $\gamma(M)$, or $\gamma_A(M)$, denote the image of (M) in $K(A)$.

(i) Show that $K(A)$ has the following universal property: for each additive function λ on the class of finitely generated A-modules, with values in an abelian group G, there exists a unique homomorphism $\lambda_0 : K(A) \to G$ such that $\lambda(M) = \lambda_0(\gamma(M))$ for all M.

(ii) Show that $K(A)$ is generated by the elements $\gamma(A/\mathfrak{p})$, where $\mathfrak{p}$ is a prime ideal of A. [Use Exercise 18.]

(iii) If A is a field, or more generally if A is a principal ideal domain, then $K(A) \cong \mathbf{Z}$.

(iv) Let $f : A \to B$ be a *finite* ring homomorphism. Show that restriction of scalars gives rise to a homomorphism $f_! : K(B) \to K(A)$ such that $f_!(\gamma_B(N)) = \gamma_A(N)$ for a B-module N. If $g : B \to C$ is another finite ring homomorphism, show that $(g \circ f)_! = f_! \circ g_!$.

Solution. (i) *Existence.* Let λ be additive: $\lambda(M') - \lambda(M) + \lambda(M'') = 0$ for every short exact sequence of finitely generated modules. $\lambda(0) = 0$ (apply additivity to $0 \to 0 \to 0 \to 0 \to 0$), and isomorphic modules $M \cong M_1$ have the same value (apply it to $0 \to M \to M_1 \to 0 \to 0$). So λ is a function on $F(A)$ and extends uniquely to a homomorphism $\tilde{\lambda} : C \to G$, $\tilde{\lambda}((M)) = \lambda(M)$. By additivity $\tilde{\lambda}$ kills each generator $(M') - (M) + (M'')$ of D, hence factors through $\lambda_0 : K(A) = C/D \to G$ with $\lambda_0(\gamma(M)) = \lambda(M)$. *Uniqueness.* $K(A)$ is generated by the elements $\gamma(M)$, on which λ_0 is prescribed.

(ii) Let M be a finitely generated module. By Ex. 18 there is a chain $0 = M_0 \subset M_1 \subset \cdots \subset M_n = M$ with $M_i/M_{i-1} \cong A/\mathfrak{p}_i$, $\mathfrak{p}_i$ prime. The exact sequences $0 \to M_{i-1} \to M_i \to M_i/M_{i-1} \to 0$ give $\gamma(M_i) = \gamma(M_{i-1}) + \gamma(A/\mathfrak{p}_i)$ in $K(A)$, so by induction $\gamma(M) = \sum_{i=1}^n \gamma(A/\mathfrak{p}_i)$. Hence the $\gamma(A/\mathfrak{p})$ generate $K(A)$.

(iii) Let A be a principal ideal domain (a field is the case with no non-zero primes) and K its field of fractions. By (ii) $K(A)$ is generated by $\gamma(A)$ and the $\gamma(A/\mathfrak{p})$ with $\mathfrak{p} = (p) \neq 0$, p a prime element. The sequence $0 \to A \xrightarrow{p} A \to A/(p) \to 0$ is exact (as A is a domain), so $\gamma(A/(p)) = \gamma(A) - \gamma(A) = 0$. Therefore $K(A)$ is generated by the single element $\gamma(A)$. The

function $\lambda(M) = \dim_K(K \otimes_A M)$ is additive (since $K = S^{-1}A$ is flat, 3.6) with values in $\mathbf{Z}$, and $\lambda(A) = 1$. By (i) there is $\lambda_0 : K(A) \to \mathbf{Z}$ with $\lambda_0(\gamma(A)) = 1$. The homomorphism $\mathbf{Z} \to K(A)$, $n \mapsto n\gamma(A)$, is surjective, and composing with λ_0 gives the identity of $\mathbf{Z}$, so it is also injective. Hence $K(A) \cong \mathbf{Z}$.

(iv) Let $f : A \to B$ be finite; B is then Noetherian (a finitely generated module over A, or a quotient of one), so $K(B)$ is defined. If N is a finitely generated B-module, it is finitely generated over A (2.16). Restriction of scalars is exact, so $N \mapsto \gamma_A(N) \in K(A)$ is an additive function on finitely generated B-modules. By (i) it induces $f_! : K(B) \to K(A)$ with $f_!(\gamma_B(N)) = \gamma_A(N)$. If $g : B \to C$ is another finite homomorphism and P is a finitely generated C-module, then $(g\circ f)_!(\gamma_C(P)) = \gamma_A(P)$, where P is an A-module via $g\circ f$, and $f_!g_!(\gamma_C(P)) = f_!(\gamma_B(P)) = \gamma_A(P)$, since regarding P as a B-module via g and then as an A-module via f is the same as regarding it as an A-module via $g \circ f$. The two homomorphisms agree on the generators $\gamma_C(P)$ of $K(C)$, so $(g \circ f)_! = f_! \circ g_!$. □

Exercise 27

Problem. Let A be a Noetherian ring and let $F_1(A)$ be the set of all isomorphism classes of finitely generated *flat* A-modules. Repeating the construction of Exercise 26 we obtain a group $K_1(A)$. Let $\gamma_1(M)$ denote the image of (M) in $K_1(A)$.

(i) Show that tensor product of modules over A induces a commutative ring structure on $K_1(A)$, such that $\gamma_1(M) \cdot \gamma_1(N) = \gamma_1(M \otimes N)$. The identity element of this ring is $\gamma_1(A)$.

(ii) Show that tensor product induces a $K_1(A)$-module structure on the group $K(A)$, such that $\gamma_1(M) \cdot \gamma(N) = \gamma(M \otimes N)$.

(iii) If A is a (Noetherian) local ring, then $K_1(A) \cong \mathbf{Z}$.

(iv) Let $f : A \to B$ be a ring homomorphism, B being Noetherian. Show that extension of scalars gives rise to a ring homomorphism $f^! : K_1(A) \to K_1(B)$ such that $f^!(\gamma_1(M)) = \gamma_1(B\otimes_A M)$. [If M is flat and finitely generated over A, then $B\otimes_A M$ is flat and finitely generated over B.] If $g : B \to C$ is another ring homomorphism (with C Noetherian), then $(f \circ g)^! = f^! \circ g^!$.

(v) If $f : A \to B$ is a finite ring homomorphism then $f_!(f^!(x)y) = xf_!(y)$ for $x \in K_1(A)$, $y \in K(B)$. In other words, regarding $K(B)$ as a $K_1(A)$-module by restriction of scalars, the homomorphism $f^!$ is a $K_1(A)$-module homomorphism.

Remark. Since $F_1(A)$ is a subset of $F(A)$ we have a group homomorphism $\epsilon : K_1(A) \to K(A)$, given by $\epsilon(\gamma_1(M)) = \gamma(M)$. If the ring A is finite-dimensional and *regular*, i.e., if all its local rings $A_\mathfrak{p}$ are regular (Chapter 11) it can be shown that ϵ is an isomorphism.

Solution. *Preliminaries.* For finitely generated flat M, N: $M \otimes N$ is finitely generated (2.12 remark) and flat (Ch. 2, Ex. 8(i)). The universal property of Ex. 26(i) holds verbatim for $K_1(A)$ and additive functions on finitely generated flat modules (same proof). If $0 \to M' \to M \to M'' \to 0$ is exact and M'' is flat, then $\mathrm{Tor}_1(M'', N) = 0$ for every N (Ch. 2, Ex. 24), so $0 \to M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ is exact for every module N; and if N is flat, tensoring any exact sequence with N gives an exact sequence.

(i) Let C_1 be the free abelian group on $F_1(A)$. Define a product on C_1 by $(M)(N) = (M\otimes N)$ on the basis and extending $\mathbf{Z}$-bilinearly; it is well defined on isomorphism classes, commutative and

associative by the canonical isomorphisms of 2.14, with identity (A). So C_1 is a commutative ring. The subgroup D_1 generated by the elements $(M')-(M)+(M'')$ for short exact sequences of finitely generated flat modules is an ideal: for N finitely generated flat, tensoring with N gives an exact sequence $0 \to M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ of finitely generated flat modules, so $((M')-(M)+(M'')) \cdot (N) = (M' \otimes N) - (M \otimes N) + (M'' \otimes N) \in D_1$. Hence $K_1(A) = C_1/D_1$ is a commutative ring with $\gamma_1(M)\gamma_1(N) = \gamma_1(M \otimes N)$ and identity $\gamma_1(A)$.

(ii) Consider the map $F_1(A) \times F(A) \to K(A)$, $((M),(N)) \mapsto \gamma(M \otimes N)$, and extend it $\mathbf{Z}$-bilinearly to $C_1 \times C \to K(A)$. It vanishes on $D_1 \times C$: for $0 \to M' \to M \to M'' \to 0$ exact with flat terms, $0 \to M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ is exact for every N (preliminaries), so $\gamma(M' \otimes N) - \gamma(M \otimes N) + \gamma(M'' \otimes N) = 0$. It vanishes on $C_1 \times D$: for $0 \to N' \to N \to N'' \to 0$ exact and M flat, $0 \to M \otimes N' \to M \otimes N \to M \otimes N'' \to 0$ is exact, so the same alternating sum is 0. Hence we get a bilinear map $K_1(A) \times K(A) \to K(A)$ with $\gamma_1(M)\gamma(N) = \gamma(M \otimes N)$. The module axioms hold on generators: $(\gamma_1(M)\gamma_1(M'))\gamma(N) = \gamma((M \otimes M') \otimes N) = \gamma_1(M)(\gamma_1(M')\gamma(N))$ by associativity of $\otimes$ and $\gamma_1(A)\gamma(N) = \gamma(A \otimes N) = \gamma(N)$.

(iii) Let A be local with maximal ideal $\mathfrak{m}$, residue field k. *Finitely generated flat modules are free.* Let M be finitely generated flat; choose $x_1, \ldots, x_n \in M$ whose images form a basis of $M/\mathfrak{m}M$. By Nakayama (2.8) they generate M; let $\phi : A^n \to M$ be the surjection and $N = \operatorname{Ker} \phi$, finitely generated since A is Noetherian. As M is flat, $\operatorname{Tor}_1(k, M) = 0$ (Ch. 2, Ex. 24), so $0 \to k \otimes N \to k^n \to k \otimes M \to 0$ is exact; $k^n \to k \otimes M = M/\mathfrak{m}M$ is an isomorphism by the choice of the x_i, so $k \otimes N = 0$, i.e. $N = \mathfrak{m}N$ and $N = 0$ by Nakayama. So $M \cong A^n$. Thus $F_1(A) = \{(A^n) : n \geq 0\}$. The sequence $0 \to A \to A^n \to A^{n-1} \to 0$ is exact, so $\gamma_1(A^n) = \gamma_1(A) + \gamma_1(A^{n-1})$ and $K_1(A)$ is generated by $\gamma_1(A)$. The rank $A^n \mapsto n$ is well defined ($A \neq 0$, Ch. 2, Ex. 11) and additive on short exact sequences of finitely generated free modules (such a sequence splits, A^n being free, so $p = m + n$ in $0 \to A^m \to A^p \to A^n \to 0$). By (the analogue of) Ex. 26(i) it induces $\lambda_0 : K_1(A) \to \mathbf{Z}$ with $\lambda_0(\gamma_1(A)) = 1$. As in Ex. 26(iii), $n \mapsto n\gamma_1(A)$ is an isomorphism $\mathbf{Z} \to K_1(A)$. So $K_1(A) \cong \mathbf{Z}$.

(iv) If M is finitely generated and flat over A, then $B \otimes_A M$ is finitely generated (2.17) and flat over B (Ch. 2, Ex. 20). If $0 \to M' \to M \to M'' \to 0$ is exact with flat terms, then $0 \to B \otimes M' \to B \otimes M \to B \otimes M'' \to 0$ is exact (M'' flat, preliminaries). So $M \mapsto \gamma_1(B \otimes_A M)$ is additive and induces $f^! : K_1(A) \to K_1(B)$ with $f^!(\gamma_1(M)) = \gamma_1(B \otimes_A M)$. It is a ring homomorphism: $f^!(1) = \gamma_1(B)$ and, using $B \otimes_A (M \otimes_A N) \cong (B \otimes_A M) \otimes_B (B \otimes_A N)$ (2.14, 2.15), $f^!(\gamma_1(M)\gamma_1(N)) = f^!(\gamma_1(M))f^!(\gamma_1(N))$. For $g : B \to C$, $C \otimes_A M \cong C \otimes_B (B \otimes_A M)$ (2.15), so the composite homomorphism $A \to B \to C$ induces $g^! \circ f^!$ on generators $\gamma_1(M)$: $(g \circ f)^! = g^! \circ f^!$. (The printed formula $(f \circ g)^! = f^! \circ g^!$ is written with the composite in the diagrammatic order: it is the statement that extension of scalars along $A \to B \to C$ is the composite of the two extensions.)

(v) Let $f : A \to B$ be finite (so B is Noetherian). It suffices to check the identity on generators $x = \gamma_1(M)$ (M finitely generated flat over A) and $y = \gamma_B(N)$ (N finitely generated over B), because both sides are biadditive in (x, y). By (ii) for B, $f^!(x)y = \gamma_1(B \otimes_A M)\gamma_B(N) = \gamma_B((B \otimes_A M) \otimes_B N)$, so $f_!(f^!(x)y) = \gamma_A((B \otimes_A M) \otimes_B N)$. By 2.14 and 2.15, $(B \otimes_A M) \otimes_B N \cong (M \otimes_A B) \otimes_B N \cong M \otimes_A (B \otimes_B N) \cong M \otimes_A N$ as A-modules. And $xf_!(y) = \gamma_1(M)\gamma_A(N) = \gamma_A(M \otimes_A N)$ by (ii) for A. These agree. $\square$

Notes

Chapter 7 has no numbered in-text exercises, and 27 end-of-chapter exercises (all solved). In Exercise 3 irreducible ideals are taken to be proper (otherwise A itself would satisfy (ii) and (iii) without being primary). In Exercise 19 I proved the stronger counting formula $(*)$ rather than the hint's replacement property, which is only sketched in the closing remark. Exercises 23 and 24 depend on the key lemma Ch. 5, Ex. 21, quoted without proof here.